MEDICAL HEALTH DATA ANALYSIS USING PYTHON

A Data Analytics Project Report

Submitted by	_____
Programme / Class	BBA
Institution	_____
Academic session	2026–2027
Dataset	Synthetic educational dataset

Academic use only. All patient-like records are simulated and contain no real personal health information.

Project Structure

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1. Objective

To perform exploratory data analysis on a synthetic health dataset using Python. The project demonstrates importing data, checking data quality, calculating descriptive statistics, grouping records, visualizing distributions, and examining correlations.

2. Software Required

• Python 3

• Jupyter Notebook

• Pandas and NumPy

• Matplotlib

• Microsoft Excel or compatible spreadsheet software

3. Introduction

Data analysis is the process of inspecting, cleaning, transforming, and summarizing information to identify patterns and communicate findings. In healthcare-related learning exercises, structured measurements such as age, body mass index (BMI), blood pressure, glucose, and cholesterol can be used to demonstrate statistical summaries and visual exploration.

This project is an educational demonstration only. The dataset is entirely synthetic. It does not describe actual people, establish medical risk, or provide a basis for diagnosis or treatment. Observed associations must not be interpreted as causal relationships.

4. Dataset Description

The dataset contains 300 simulated records and 13 columns. Each row represents one artificial record identified by a synthetic ID. No names, addresses, contact details, or real patient identifiers are included.

Patient_ID	Age_years	Gender	Height_m	Weight_kg	BMI	Systolic_BP_mmHg
SYN0001	23	Male	1.76	66.6	21.5	108
SYN0002	66	Male	1.71	77.2	26.4	120
SYN0003	59	Female	1.45	67.0	31.7	133
SYN0004	45	Male	1.79	71.6	22.2	108
SYN0005	45	Female	1.55	70.1	29.1	119

Variables include age, gender, height, weight, BMI, systolic and diastolic blood pressure, glucose, total cholesterol, smoking status, activity level, and a descriptive BMI group.

5. Procedure

• Step 1: Import Pandas, NumPy, and Matplotlib in the Jupyter Notebook.

• Step 2: Load the accompanying CSV dataset using pandas.read_csv().

• Step 3: Inspect the first rows, dataset dimensions, column types, and summary statistics.

• Step 4: Check missing values and duplicate records to assess basic data quality.

• Step 5: Calculate descriptive measures such as mean, median, minimum, and maximum.

• Step 6: Group records by gender, BMI category, and activity level.

• Step 7: Create charts to explore age, BMI, blood pressure, glucose, and cholesterol patterns.

• Step 8: Calculate a correlation matrix for selected numeric variables.

• Step 9: Interpret the charts cautiously and document the findings and limitations.

6. Output

The analysis generated the following visual outputs. Each chart is based on the synthetic dataset supplied with this report.

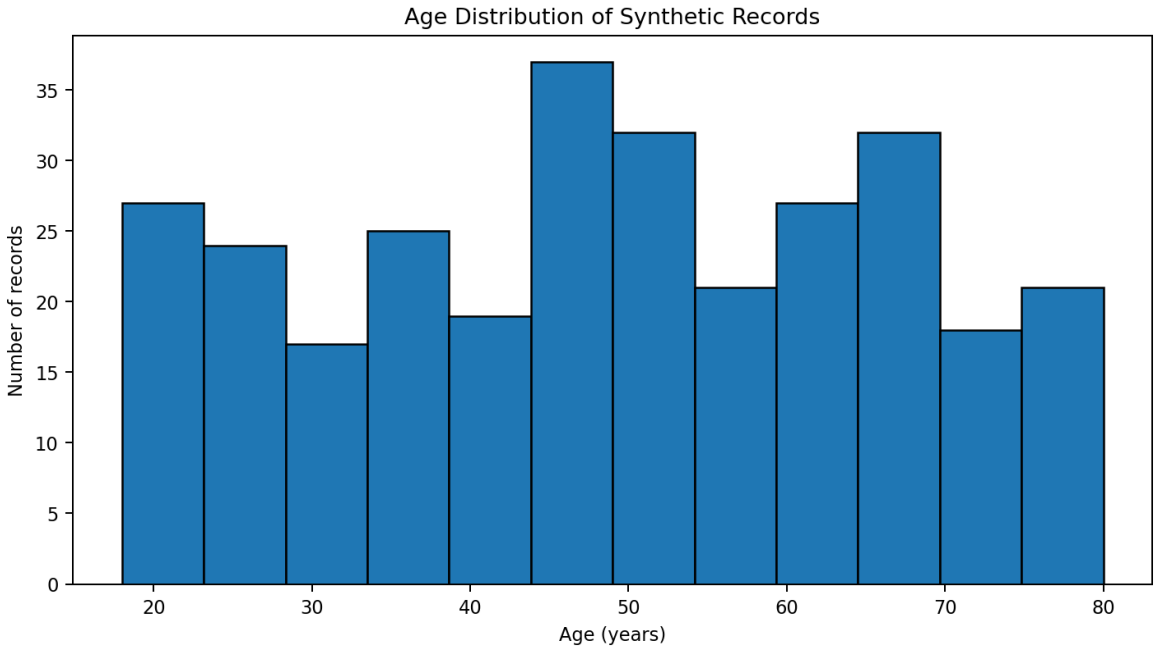


Figure 1. Age distribution: the histogram summarizes the number of synthetic records in each age interval.

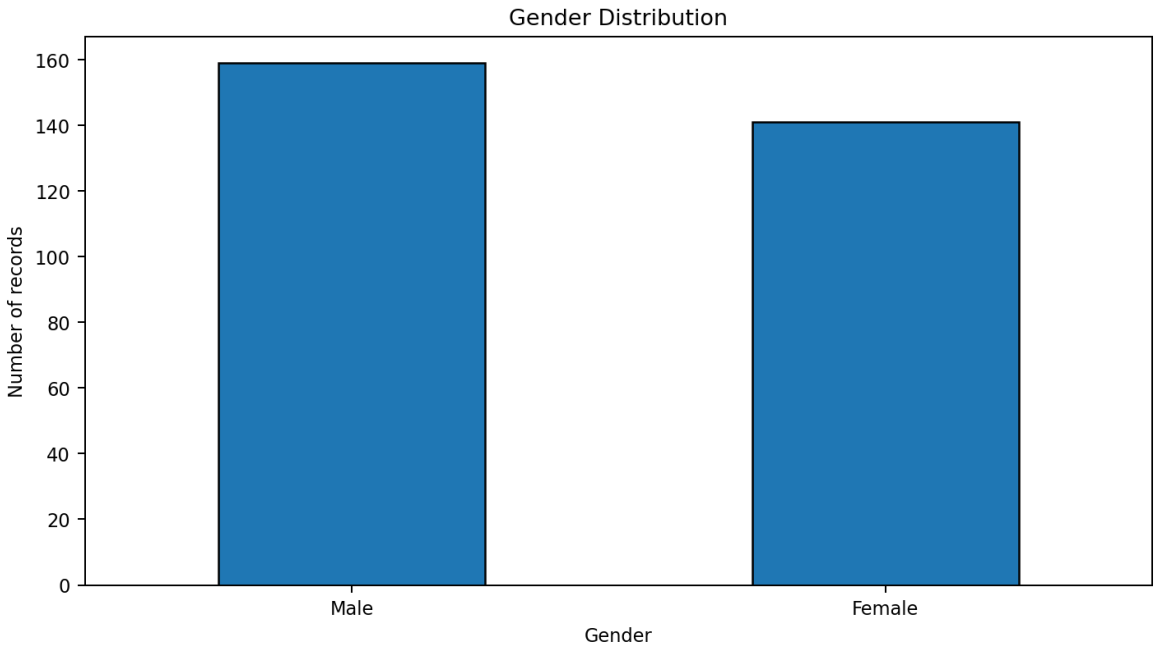


Figure 2. Gender distribution: counts for the two simulated gender categories.

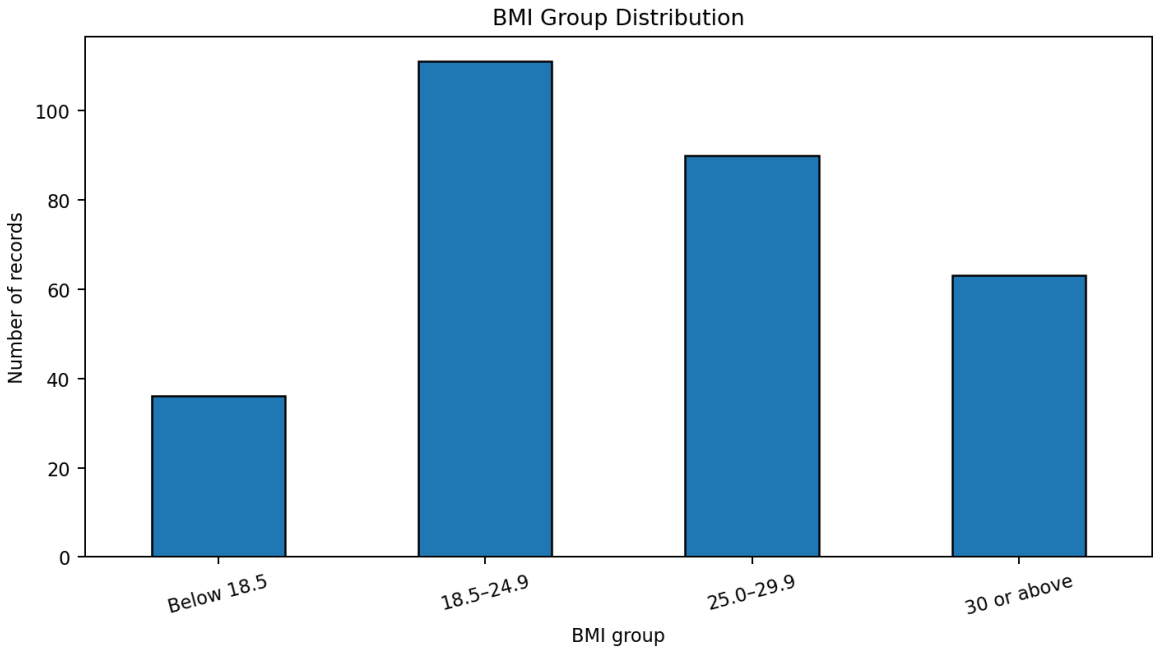


Figure 3. BMI group distribution: descriptive group counts using standard numeric cut-points. These group labels are not a diagnosis.

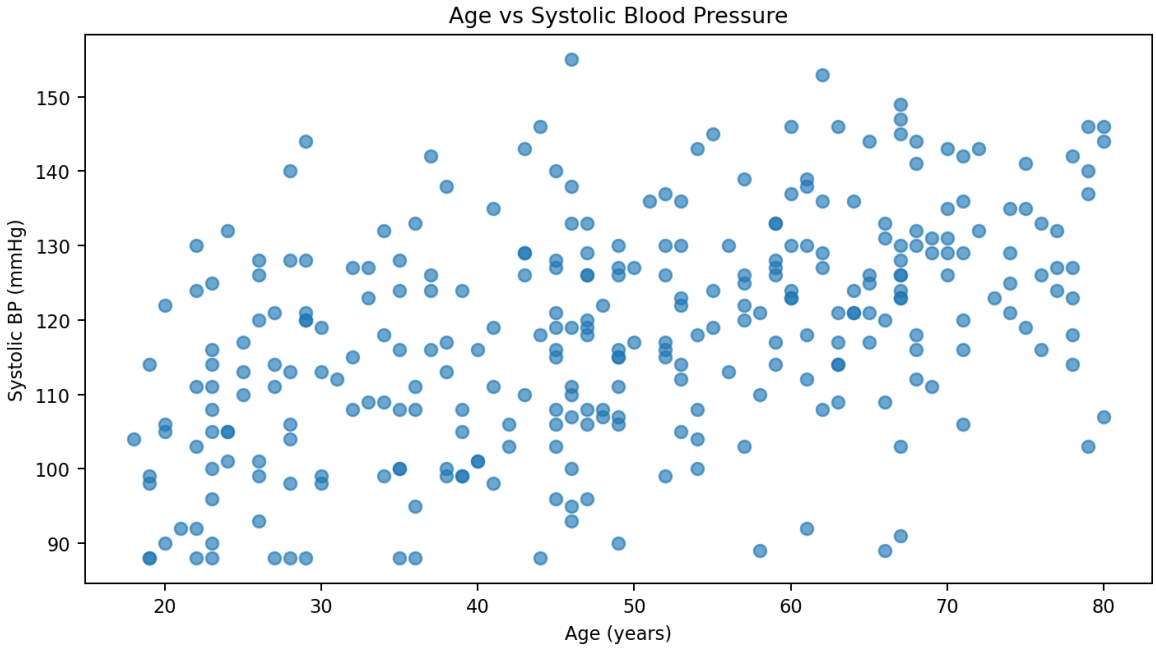


Figure 4. Age versus systolic blood pressure: each point represents one synthetic record.

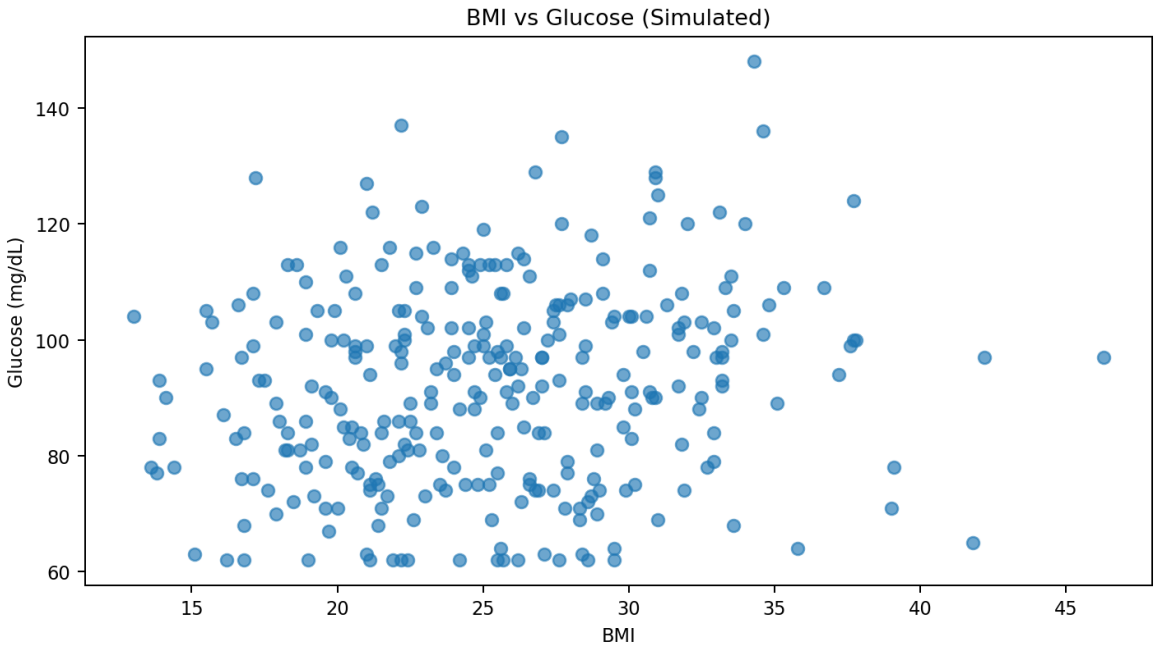


Figure 5. BMI versus glucose: exploratory scatter plot; no clinical inference should be drawn.

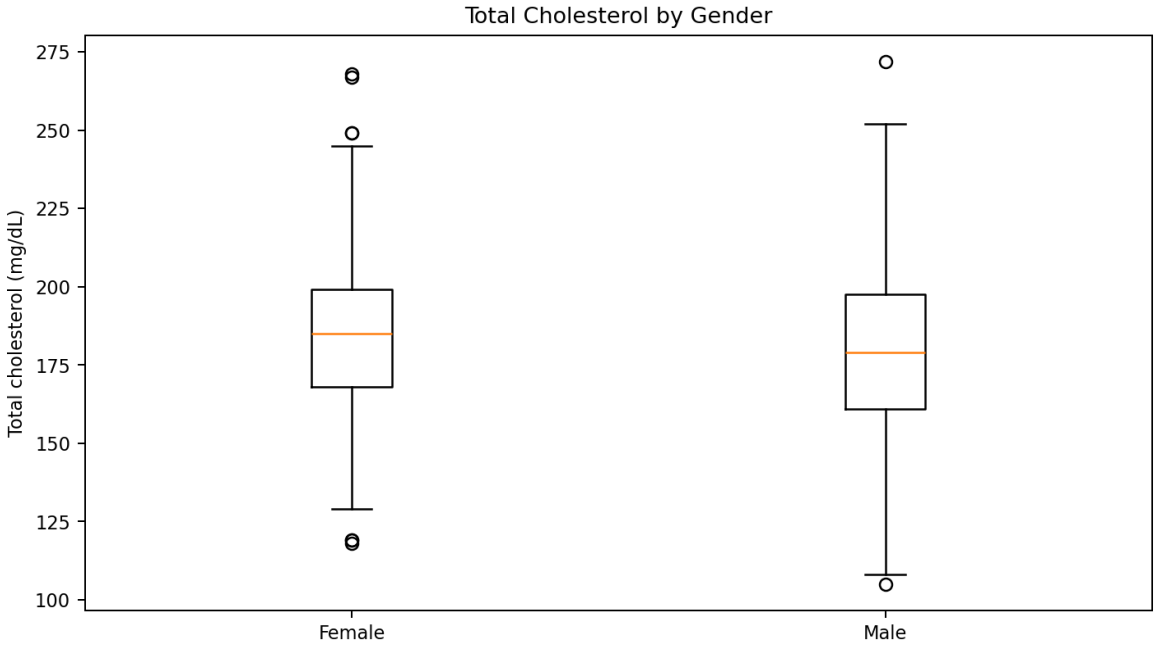


Figure 6. Total cholesterol by gender: box plots show median and spread in the simulated groups.

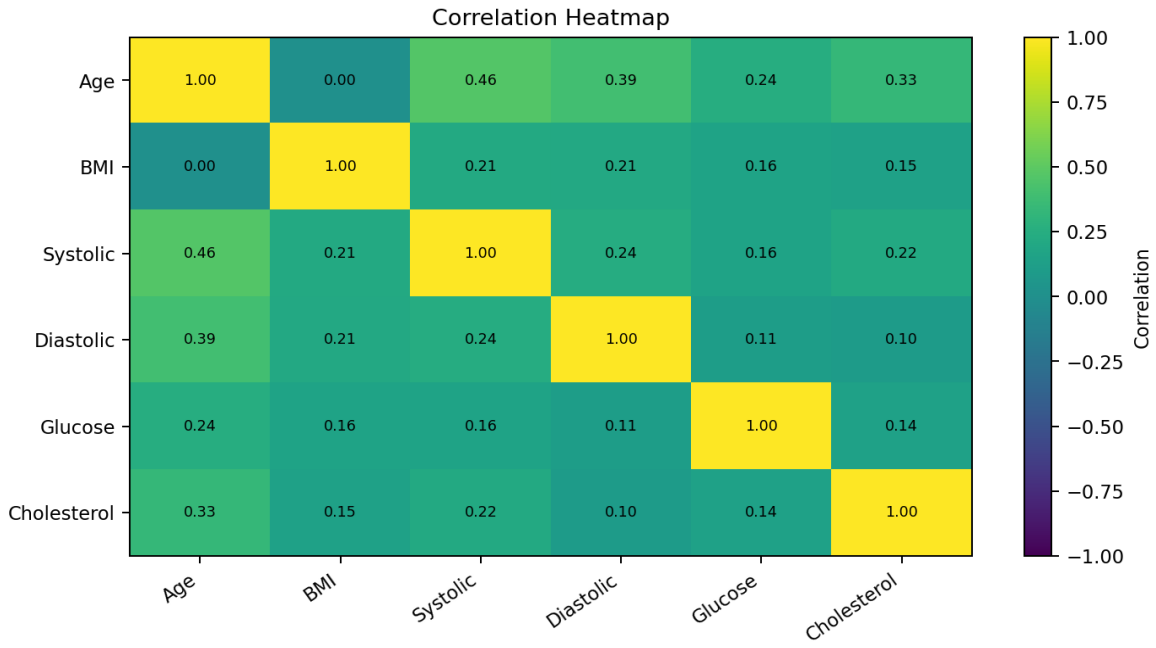


Figure 7. Correlation heatmap: values range from -1 to +1 and summarize linear association in this dataset.

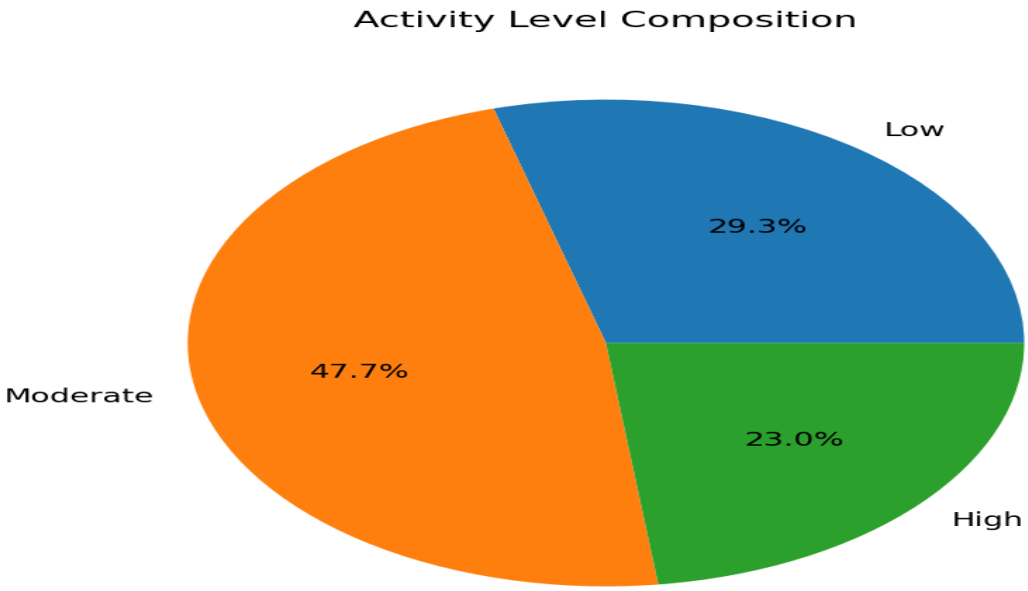


Figure 8. Activity level composition across synthetic records.

7. Observation

Measure	Dataset result
Number of records	300
Average age (years)	49.0
Average BMI	25.3
Average systolic BP (mmHg)	118.3
Average diastolic BP (mmHg)	74.7
Average glucose (mg/dL)	92.0
Average total cholesterol (mg/dL)	181.2

The age histogram shows how the simulated sample is distributed across age bands. Categorical charts summarize the generated gender, BMI-group, and activity-level counts. Scatter plots allow visual inspection of possible relationships between selected measurements. The correlation matrix provides a compact summary of pairwise linear associations.

Because values were generated using a simulation process, any apparent pattern is a property of the synthetic data-generation assumptions and random variation. It cannot be generalized to a real population.

8. Result

Successfully completed an exploratory data analysis workflow using Python and a synthetic health dataset. The project demonstrates data loading, data-quality checks, descriptive statistics, grouping, chart creation, and correlation analysis. The notebook, CSV file, Excel appendix, and visual charts are supplied as supporting project materials.

Limitations and Ethical Note

This dataset is synthetic and created solely for classroom practice. It is not sourced from a hospital or patient record system. It should not be used for diagnosis, treatment, screening, clinical decisions, or claims about real populations. BMI categories and measured values are presented for statistical practice only.

Appendix A – Dataset Sample

The complete dataset is supplied separately as a CSV file and as an Excel workbook. The workbook contains a Synthetic_Data sheet and a Data_Dictionary sheet. A short sample is shown below.

Patient_ID	Age_years	Gender	Height_m	Weight_kg	BMI
SYN0001	23	Male	1.76	66.6	21.5
SYN0002	66	Male	1.71	77.2	26.4
SYN0003	59	Female	1.45	67.0	31.7
SYN0004	45	Male	1.79	71.6	22.2
SYN0005	45	Female	1.55	70.1	29.1
SYN0006	72	Male	1.68	80.8	28.7
SYN0007	23	Female	1.51	76.3	33.3
SYN0008	61	Male	1.67	48.5	17.5

Appendix B – Data Dictionary

Column	Meaning
Patient_ID	Synthetic record code
Age_years	Age in years
Gender	Simulated category
Height_m / Weight_kg	Height in metres / weight in kilograms
BMI	Weight divided by height squared
Systolic_BP_mmHg / Diastolic_BP_mmHg	Simulated blood pressure values
Glucose_mg_dL	Simulated glucose value in mg/dL
Total_Cholesterol_mg_dL	Simulated total cholesterol in mg/dL
Smoking_Status / Activity_Level	Simulated categorical variables
BMI_Group	Descriptive BMI band